Bounded flat transversals for fat convex bodies

Kaiwen Jin*

Abstract

Fix $d \geq 3$, $1 \leq k \leq d-2$, $p \geq k+2$, and a fatness parameter $\rho \geq 1$. We prove that a finite family of ρ-fat compact convex bodies in $\mathbb{R}^d$ can be intersected by a bounded number of affine k-flats whenever every p members contain $k+2$ members with a common k-flat transversal. The bound depends only on d, k, p, ρ; the bodies may have arbitrary sizes and convex boundaries. The missing rounding step is obtained by grouping bodies around members of minimum radius, then reducing the dimension of their transversals on the sphere. A finite family of perturbations turns the resulting approximate transversals into exact ones at the scale of the smallest body in each group.

1 Introduction

Geometric transversal questions ask how a condition on small subfamilies controls the number of probes needed to reach an entire family. For affine subspaces of intermediate dimension, the shape of the sets matters. Here we consider convex bodies with a fixed bound on their eccentricity but with no restriction on their individual sizes.

A compact convex body $K \subset \mathbb{R}^d$ is ρ-fat if there are a center x_K and positive radii r_K, R_K such that

$$B(x_K, r_K) \subseteq K \subseteq B(x_K, R_K), \qquad R_K \leq \rho r_K. \tag{1}$$

All balls are closed. The radii in (1) are allowed to vary from one body to another. A k-flat is a k-dimensional affine subspace, and a k-transversal of a subfamily is one k-flat meeting every member of that subfamily.

Jung and Pálvölgyi proved a fractional Helly theorem for k-transversals of fat convex sets [2, Theorem 5]. Combined with the fractional transversal theorem of Alon, Kalai, Matoušek and Meshulam [1], in the published formulation of [2, Theorem 22], this bounds the fractional number of probes under the local condition considered below. Jung and Pálvölgyi also proved the corresponding integer bound for balls and for families having common inner and outer radius bounds [2, Theorem 7 and Corollary 8]. Their Conjecture 1 asks for the extension to general fat convex sets. The remaining question is whether a fractional cover can be replaced by a bounded family of flats when the radii vary freely.

Theorem 1.1. *Let $d \geq 3$, $1 \leq k \leq d-2$, $p \geq k+2$ be integers, and let $\rho \geq 1$. There is a constant $C = C(d, k, p, \rho)$ with the following property. If $\mathcal{F}$ is a finite family of ρ-fat compact convex bodies in $\mathbb{R}^d$, and every p members of $\mathcal{F}$ contain $k+2$ members with a common k-transversal, then at most C k-flats together intersect every member of $\mathcal{F}$.*

*yui.ui.siki@gmail.com

Thus the theorem gives the finite-family form of [2, Conjecture 1], with the same fatness convention and local condition. The constant is independent of the number of bodies, their locations, their global size ratio, and their boundary description complexity.

The proof uses the fractional cover only at the end. We first prove a uniform weak-net statement for a finite family of bodies that are each hit with positive probability by a finite distribution of flats. Bodies are grouped by their overlap with a body of minimum outer radius. There are boundedly many groups, and within each group a fixed positive amount of probe mass hits both the smallest body and any specified member. Moving those probes through the center of the smallest body allows a radial projection and decreases the dimension of the transversals.

The movement introduces an absolute error equal to the radius of that smallest body. Every other member of the group is at least as large. Convexity and fatness then give a ball of comparable radius near each approximate probe, so a fixed finite collection of perturbations reaches all the original bodies. On the sphere this correction is made in finitely many gnomonic charts. The argument retains tangential incidences and allows new probes outside the support of the original distribution. The resulting bounds are not optimized.

Section 2 gives the grouping and geometric facts. Section 3 proves the spherical and affine weak-net statements and then Theorem 1.1.

2 Grouping and geometric facts

All probability measures below have finite support, and expectations are finite weighted sums. Families are indexed, so labels are retained when bodies or projected images coincide.

2.1 Grouping sets of positive measure

The following elementary observation lets us use one smallest scale within each group.

Lemma 2.1. *Let $A_1, \dots, A_n$ be measurable sets in a finite probability space with measure μ, and suppose that $\mu(A_i) \geq \varepsilon > 0$. Assign a positive number R_i to each label. There is a partition of the labels into at most*

$$M(\varepsilon) = \lceil 2/\varepsilon^2 \rceil$$

groups, with a distinguished label i_ℓ in each group I_ℓ, such that, for every $i \in I_\ell$,

$$R_i \geq R_{i_\ell}, \qquad \mu(A_i \cap A_{i_\ell}) \geq \delta, \qquad \delta = \varepsilon^2/2.$$

Proof. Choose a label of minimum R_i among those remaining and remove as its group every remaining label whose overlap with it is at least δ. Repeat, breaking ties arbitrarily. The minimum-radius condition is immediate, and every chosen label removes itself.

If $i_1, \dots, i_t$ are the chosen labels, then their pairwise overlaps are less than δ. For $Z = \sum_{\ell=1}^{t} 1_{A_{i_\ell}}$, the nonnegativity of its variance gives

$$t^2\varepsilon^2 \leq (\mathbb{E}_\mu Z)^2 \leq \mathbb{E}_\mu Z^2 \leq t + t(t-1)\delta.$$

Hence $t(\varepsilon^2 - \delta) \leq 1 - \delta$, which implies $t \leq 2/\varepsilon^2$. □

Conditioning μ on A_{i_ℓ} is legitimate because its measure is positive. In that conditional probability, every group member has overlap mass at least $\delta/\mu(A_{i_\ell}) \geq \delta$. The grouping itself always uses the original measure.

2.2 Perturbing an affine transversal

An approximate transversal at one fixed absolute scale can be replaced by a finite family of exact transversals.

Lemma 2.2. *Fix $0 \le j < m$, $\sigma \ge 1$, $a \ge 0$, $s > 0$, and an affine j-flat $F \subset \mathbb{R}^m$. Every family of σ-fat compact convex bodies with specified outer radii $R_K \ge s$ and $\operatorname{dist}(K, F) \le as$ can be hit by at most*

$$Q(m-j, \sigma, a), \qquad Q(b, \sigma, a) = \left(2\lceil 4\sigma\sqrt{b}(a+1)\rceil + 1\right)^b \tag{2}$$

j-flats parallel to F.

Proof. Translate the origin onto F, and project orthogonally onto its normal space of dimension $b = m - j$. Each image A is convex and has the same concentric inner and outer radius bounds as its original body. Choose $y \in A$ with $|y| \le as$, and let c be the projected center. With $\lambda = s/(2R_K)$ and $z = (1-\lambda)y + \lambda c$, convexity gives

$$B(z, \lambda r_K) \subseteq A, \qquad \lambda r_K \ge \frac{s}{2\sigma}, \qquad |z - y| \le \frac{s}{2}.$$

Here $\lambda \le 1/2$, and $y \in B(c, R_K)$. Thus $|z| \le (a + 1/2)s$.

In an orthonormal coordinate system on the normal space, take grid points with spacing $h = s/(4\sigma\sqrt{b})$ and coordinate indices $|q_\nu| \le \lceil (a+1)s/h \rceil$. A nearest grid point to z lies in this finite grid and has distance at most $\sqrt{b}\, h/2 = s/(8\sigma)$ from z. It therefore belongs to A. There are at most $Q(b, \sigma, a)$ grid points, and the corresponding parallel flats hit every original body. □

The lower bound $R_K \ge s$ is essential to this use of a common grid. The error in the lemma is as, not aR_K.

2.3 Spherical projections

On the unit sphere $\mathbb{S}^m \subset \mathbb{R}^{m+1}$, distances and cap radii are geodesic. A proper spherical convex body is a compact set with positive inradius, contained in an open hemisphere and closed under the shorter great-circle arcs. We also allow the whole sphere. Spherical σ-fatness means concentric cap bounds

$$B(x_K, r_K) \subseteq K \subseteq B(x_K, R_K), \qquad 0 < r_K \le R_K \le \sigma r_K, \quad R_K \le \pi.$$

A great j-sphere is the intersection with a linear $(j+1)$-dimensional subspace. For $j = 0$ it is an antipodal pair. We fix

$$\gamma = \pi/100.$$

For $w \in \mathbb{S}^m$, the gnomonic map to $P_w = \{z : w \cdot z = 1\}$ is

$$g_w(u) = \frac{u}{w \cdot u}, \qquad g_w^{-1}(w + x) = \frac{w + x}{\sqrt{1 + |x|^2}}, \quad x \perp w. \tag{3}$$

It is defined on the open hemisphere centered at w and maps great-circle arcs to line segments. Its inverse has differential norm at most 1. On $B(w, \pi/4)$, the differential of g_w has radial factor $\sec^2 t$ and transverse factor $\sec t$, where $t = \operatorname{dist}(u, w)$; both are at most 2. Since this cap is geodesically convex,

$$\operatorname{dist}(g_w^{-1}(z), g_w^{-1}(z')) \le |z - z'|, \qquad |g_w(u) - g_w(v)| \le 2 \operatorname{dist}(u, v) \tag{4}$$

in the respective domains. Consequently a σ-fat body contained in that cap has a convex image with concentric Euclidean radii $r_K, 2R_K$, and is 2σ-fat.

Lemma 2.3. *Suppose $R_K < \gamma$ and $0 < s \le R_K$, and let K^s be the closed s-neighborhood of K on the sphere. Then $D = \mathrm{conv}_{\mathbb{S}}(K^s)$ is a compact σ-fat spherical body, with concentric radii $r_K + s, R_K + s$, and*

$$D \subseteq K^{2s}.$$

Proof. The neighborhood contains $B(x_K, r_K + s)$ and is contained in $B(x_K, R_K + s)$. The latter cap has radius below $2\gamma < \pi/4$ and is convex, so it contains D. Moreover

$$\frac{R_K + s}{r_K + s} \le \frac{R_K}{r_K} \le \sigma.$$

Use g_{x_K} on this cap. Each point of $g_{x_K}(K^s)$ is within $2s$ of the convex set $g_{x_K}(K)$, by (4). Taking convex hulls gives

$$g_{x_K}(D) \subseteq g_{x_K}(K) + B(0, 2s).$$

The inverse bound in (4) proves $D \subseteq K^{2s}$. The Euclidean convex hull in this chart is compact, which also proves compactness of D. □

Lemma 2.4. *Let $D \subset \mathbb{S}^m$ be a σ-fat spherical convex body with specified outer radius $R < \pi/4$, and let $\omega \in \mathbb{S}^m$. If D avoids both ω and $-\omega$, then*

$$\Pi_\omega(y) = \frac{y - (y \cdot \omega)\omega}{|y - (y \cdot \omega)\omega|}$$

maps D to a compact spherical convex $\pi^2\sigma$-fat body in the equatorial sphere $\mathbb{S}^{m-1} \subset \omega^\perp$. Great j-spheres through ω, for $j \ge 1$, project to great $(j-1)$-spheres, and their incidences with D lift exactly.

Proof. Write x, r, R for the concentric cap data, put $t = \mathrm{dist}(x, \omega)$, $s_0 = \min(t, \pi - t)$, and $w = \Pi_\omega(x)$. Since the inner cap avoids both poles, $r < s_0$. For $u < s_0$, the image of $B(x, u)$ is the cap centered at w of radius

$$\alpha_u = \arcsin\left(\frac{\sin u}{\sin t}\right). \tag{5}$$

To see this, fix a direction $v \in \omega^\perp$ and consider $y = \sin\theta\, v + \cos\theta\, \omega$, $0 < \theta < \pi$. Maximizing $y \cdot x$ shows that this semicircle reaches $B(x, u)$ exactly when $\angle(v, w) \le \arcsin(\sin u / \sin t)$. The condition $u < s_0$ excludes the two endpoint poles.

If $R < s_0$, both projected cap bounds have the same center and

$$\frac{\alpha_R}{\alpha_r} \le \frac{\pi}{2}\frac{\sin R}{\sin r} \le \frac{\pi^2}{4}\frac{R}{r} \le \frac{\pi^2}{4}\sigma.$$

If $R \ge s_0$, then

$$\alpha_r \ge \frac{\sin r}{\sin t} \ge \frac{2r}{\pi s_0} \ge \frac{2}{\pi\sigma}.$$

Use the whole equatorial sphere as an outer cap of radius π, centered at w. Its radius ratio is at most $\pi^2\sigma/2$. Both cases give the stated conservative bound.

The image is compact because D avoids the poles. Projection preserves positive combinations, followed by normalization, so the image is spherically convex. For completeness, its Euclidean convex hull cannot contain zero: a vanishing positive combination of projected directions, after dividing the coefficients by the positive projection lengths, would lift to a nonzero positive combination of points of D whose transverse component vanishes. Normalizing that combination gives a pole in D.

It cannot be zero because D lies in one cap of radius below $\pi/4$. A minimum-norm point of the compact convex hull strictly separates it from zero, so the image lies in an open hemisphere.

If a linear $(j+1)$-space contains ω, its intersection with $\omega^\perp$ has dimension j. Conversely, spanning such a j-space with ω restores the original dimension. A point of D belongs to the lifted space precisely when its projected direction belongs to the equatorial space. □

2.4 Perturbing a great-sphere transversal

Lemma 2.5. *Fix $0 \le j < m$, $\sigma \ge 1$, $s > 0$, and a great j-sphere $F \subset \mathbb{S}^m$. Every family of σ-fat spherical bodies with*

$$s \le R_K < \gamma, \qquad \operatorname{dist}(K, F) \le 2s$$

can be intersected by at most $B(m, j, \sigma)$ great j-spheres, where B is independent of s and the family.

Proof. Fix a γ-net on F of cardinality A_j, depending only on j and the fixed γ. For a body K, choose $y \in K$, $f \in F$ with $\operatorname{dist}(y, f) \le 2s$, and a net point w with $\operatorname{dist}(f, w) \le \gamma$. Every point of K lies within

$$2R_K + 2s + \gamma < 5\gamma < \pi/4$$

of w.

In the chart g_w, F becomes an affine j-flat. The image body is 2σ-fat, has specified outer radius $2R_K \ge s$, and is within $4s$ of that affine flat. Lemma 2.2 gives at most $Q(m-j, 2\sigma, 4)$ parallel affine flats hitting all bodies assigned to this chart. Every affine j-flat in P_w spans a linear $(j+1)$-space with the origin, so its inverse image lies on a great j-sphere. These lifted spheres hit the original bodies. Thus we may take

$$B(m, j, \sigma) = A_j Q(m-j, 2\sigma, 4). \tag{6}$$

For $j = 0$, use the two points of F as its net; every chart point lifts to an antipodal pair. □

3 Weak nets and the bounded transversal theorem

3.1 Great-sphere transversals

We first prove the spherical statement needed for the affine problem. For a spherical body K, write $H_j(K)$ for the family of great j-spheres meeting it.

Theorem 3.1. *For $m \ge 1$, $0 \le j < m$, $\sigma \ge 1$, and $\varepsilon \in (0, 1]$, there is a finite $S(m, j, \sigma, \varepsilon)$ such that the following holds. If μ is a finitely supported probability measure on great j-spheres and $\mathcal{F}$ is a finite family of σ-fat spherical convex bodies satisfying $\mu(H_j(K)) \ge \varepsilon$ for all $K \in \mathcal{F}$, then at most $S(m, j, \sigma, \varepsilon)$ great j-spheres hit the whole family.*

Proof. We induct on j, for all $m > j$. Bodies with $R_K \ge \gamma$ contain caps of radius at least γ/σ, so a fixed metric net of $\mathbb{S}^m$ with mesh $\gamma/(2\sigma)$ hits all of them. Extend its points to great j-spheres. Let $G(m, \sigma)$ be the cardinality of this fixed net, and remove these large bodies.

Apply Lemma 2.1 to the remaining incidence ranges and their specified outer radii. There are at most $M(\varepsilon)$ groups. In one group, let the distinguished body have center ω and outer radius s. Every group member has $s \le R_K < \gamma$. Put $\delta = \varepsilon^2/2$.

For $j = 0$, a positive overlap means that some antipodal pair hits both the distinguished body and K. Hence $\operatorname{dist}(K, \{\omega, -\omega\}) \le s$. Lemma 2.5 covers the group by at most $B(m, 0, \sigma)$ antipodal pairs. This proves the base case.

Now let $j \geq 1$, and condition μ on probes hitting the distinguished body. Each such great sphere F has distance at most s from ω. Normalize the orthogonal projection of ω onto its linear space, obtaining $v \in F$, and rotate in $\operatorname{span}(v, \omega)$ to send v to ω, fixing the orthogonal complement. The rotation angle is at most s. If $v = \omega$, take the identity rotation. Every point moves by at most s: for a rotation of angle θ, the dot product of a unit vector with its rotated image is at least $\cos\theta$.

For a group member put $D_K = \operatorname{conv}_{\mathbb{S}}(K^s)$. Lemma 2.3 makes it σ-fat, with outer radius $R_K + s < 2\gamma$, and gives $D_K \subseteq K^{2s}$. A probe hitting both K and the distinguished body becomes a great sphere through ω hitting D_K.

Project the rotated probes onto $\omega^{\perp}$. This defines one finitely supported probability measure ν on great $(j-1)$-spheres, independently of the choice of K. For each D_K avoiding the poles, Lemma 2.4 gives a $\pi^2\sigma$-fat projected body, and

$$\nu(H_{j-1}(\Pi_\omega D_K)) \geq \frac{\mu(H_j(K) \cap H_j(K_i))}{\mu(H_j(K_i))} \geq \delta,$$

where K_i denotes the distinguished body. The induction hypothesis supplies $S(m-1, j-1, \pi^2\sigma, \delta)$ equatorial probes. Lift them through ω. They hit all the corresponding D_K, and also all D_K containing a pole. If no projected bodies remain, choose one arbitrary great j-sphere through ω.

Each K is now within $2s$ of one lifted sphere. Apply Lemma 2.5 to the subfamily assigned to each lifted sphere; all its members have $R_K \geq s$. This covers the group by boundedly many exact probes.

More explicitly, define the finite bounds recursively by

$$\begin{aligned} S(m, 0, \sigma, \varepsilon) &= G(m, \sigma) + M(\varepsilon) B(m, 0, \sigma), \\ S(m, j, \sigma, \varepsilon) &= G(m, \sigma) + M(\varepsilon) B(m, j, \sigma) \max\{1, S(m-1, j-1, \pi^2\sigma, \varepsilon^2/2)\}, \quad j \geq 1. \end{aligned} \tag{7}$$

The recursion terminates at $j = 0$. □

3.2 Affine transversals

Let $\mathcal{A}_{d,k}$ be the space of affine k-flats in $\mathbb{R}^d$, and write $H_k(K) = \{L \in \mathcal{A}_{d,k} : L \cap K \neq \varnothing\}$.

We use one radial projection fact. If a compact convex σ-fat Euclidean body avoids the origin, its radial image is a compact spherical convex $\pi\sigma$-fat body. Indeed, for its concentric data x, r, R, put $t = |x| > r$. The inner ball projects to the cap centered at x/t with radius $\arcsin(r/t)$. If $t > R$, the corresponding outer radius is $\arcsin(R/t)$, and the ratio is at most $(\pi/2)R/r$. If $t \leq R$, the inner radius is at least $r/t \geq 1/\sigma$; use an outer cap of radius π with the same center. Radial projection preserves normalized positive combinations. The image lies in an open hemisphere: a nearest point y of the body to the origin satisfies $y \cdot z \geq |y|^2 > 0$ for every point z of the body. Compactness follows from avoidance of the origin.

Theorem 3.2. *Fix $d \geq 2$, $1 \leq k < d$, $\rho \geq 1$, and $\varepsilon \in (0, 1]$. There is a finite $W(d, k, \rho, \varepsilon)$ such that, for every finitely supported probability measure μ on $\mathcal{A}_{d,k}$, every finite family of ρ-fat compact convex bodies satisfying $\mu(H_k(K)) \geq \varepsilon$ is hit by at most $W(d, k, \rho, \varepsilon)$ affine k-flats.*

Proof. Apply Lemma 2.1 to the incidence ranges with sizes R_K. For one group let the distinguished body have center a and outer radius s, and condition the measure on its hitting flats. Translate each such flat through a by the shortest displacement, whose length is at most s.

Set $D_K = K + B(0, s)$. It has concentric radii $r_K + s, R_K + s$ and remains ρ-fat. Every original probe hitting both K and the distinguished body becomes a flat through a hitting D_K. Take a as the origin and push the conditional measure to the great $(k-1)$-spheres given by these linear k-spaces.

This measure hits the radial image of each D_K avoiding a with probability at least $\delta = \varepsilon^2/2$. Those images are $\pi\rho$-fat by the radial fact just proved.

Theorem 3.1 gives at most $S(d-1, k-1, \pi\rho, \delta)$ probes on $\mathbb{S}^{d-1}$. Their linear lifts through a also hit every D_K containing a. If every D_K contains a, choose one arbitrary k-flat through it. Each original K is at distance at most s from one of these lifted flats, and all group members have $R_K \geq s$. Lemma 2.2, with $a = 1$, replaces each lifted flat by at most $Q(d-k, \rho, 1)$ exact probes. Taking all groups, a valid bound is

$$W(d, k, \rho, \varepsilon) = M(\varepsilon)Q(d-k, \rho, 1)\max\{1, S(d-1, k-1, \pi\rho, \varepsilon^2/2)\}. \tag{8}$$

□

3.3 Proof of the main theorem

The published fractional results give the following precise input. For fixed d, k, p, ρ in Theorem 1.1, there is $T = T(d, k, p, \rho) \geq 1$ such that every nonempty family under its hypotheses has finitely many flats $L_1, \ldots, L_N$ and nonnegative weights $w_1, \ldots, w_N$ satisfying

$$\sum_{\nu=1}^{N} w_\nu \leq T, \qquad \sum_{\nu : L_\nu \cap K \neq \varnothing} w_\nu \geq 1 \quad (K \in \mathcal{F}). \tag{9}$$

To obtain this, use [2, Theorem 5] on the flat-incidence ranges and [2, Theorem 22], which credits the fractional transversal theorem to Alon, Kalai, Matoušek and Meshulam [1]. In the latter theorem take $q = k + 2$. Its hypotheses are therefore the original local condition and the published fat-body fractional Helly property. Finite support in (9) causes no restriction: on a finite body family there are only finitely many realizable incidence patterns. Choose one representative flat for each pattern and collect its weights. The resulting finite cover problem attains its optimum.

Proof of Theorem 1.1. If $|\mathcal{F}| < p$, one flat through a point of each body gives at most $p - 1$ probes. Otherwise normalize the weights in (9) by their total. That total is positive, and every body has hitting probability at least $1/T$. Apply Theorem 3.2 with $\varepsilon = 1/T$. Thus a valid choice is

$$C(d, k, p, \rho) = \max\{p - 1, W(d, k, \rho, 1/T(d, k, p, \rho))\}.$$

The definitions (2), (6), (7), and (8) show that every bound is finite and depends only on the displayed fixed parameters. □

AI-use statement. This manuscript was prepared with substantial assistance from OpenAI Codex, including proof exploration, source checking, and drafting. The mathematical claims are submitted for independent review.

References

[1] N. Alon, G. Kalai, J. Matoušek, and R. Meshulam, *Transversal numbers for hypergraphs arising in geometry*, Advances in Applied Mathematics **29** (2002), no. 1, 79–101. https://doi.org/10.1016/S0196-8858(02)00003-9. Author manuscript (21 May 2001).

[2] A. Jung and D. Pálvölgyi, *k-Dimensional Transversals For Fat Convex Sets*, Discrete & Computational Geometry (2026). https://doi.org/10.1007/s00454-026-00852-8.